

H₂O Vaporization

PHASE EQUILIBRIUM EQUATION

Another important aspect is the vaporization of H₂O. It has been reported that complete vaporization of H₂O could occur during gas injection, which would in turn affect the gas injectivity.

To model H₂O vaporization, the equation for thermodynamic equilibrium [\(D2.1\)](#) is applied to the H₂O component, i.e.

$$g_{n_c} = f_{\text{H}_2\text{O},g} - f_{\text{H}_2\text{O},w} = 0 \quad (\text{D3.1})$$

The fugacity $f_{\text{H}_2\text{O},g}$ is calculated from a cubic EOS as in the case of other gaseous components. The fugacity $f_{\text{H}_2\text{O},w}$ is calculated from the procedure described below.

CALCULATION OF H₂O FUGACITY IN AQUEOUS PHASE

Calculation of Fugacity of Saturated H₂O at Reservoir Temperature T

The equation due to [Canjar and Manning_\(1967\)](#) is used:

$$\frac{f_{\text{H}_2\text{O}}^s}{p_{\text{H}_2\text{O}}^s} = \begin{cases} 0.9958 + 9.68330 \times 10^{-5}T' - 6.1750 \times 10^{-7}T'^2 - 3.08333 \times 10^{-10}T'^3, & T' > 90^\circ F \\ 1, & \text{otherwise} \end{cases} \quad (\text{D3.2})$$

where

$f_{\text{H}_2\text{O}}^s$ = fugacity of H₂O at saturation pressure and temperature T' (°F)

$p_{\text{H}_2\text{O}}^s$ = saturation pressure of H₂O at temperature T' (°F)

T' = temperature in °F corresponding to T

Calculation of Saturation Pressure of H₂O at Reservoir Temperature T

The equation due to [Saul and Wagner \(1987\)](#) is used:

$$\ln \left(\frac{p_{\text{H}_2\text{O}}^s}{p_{c,\text{H}_2\text{O}}} \right) = \frac{T_{c,\text{H}_2\text{O}}}{T} (a_1 \tau + a_2 \tau^{1.5} + a_3 \tau^3 + a_4 \tau^{3.5} + a_5 \tau^4 + a_6 \tau^{7.5}) \quad (\text{D3.3})$$

with

$$a_1 = -7.85823, \quad a_4 = +22.6705$$

$$a_2 = +1.83991, \quad a_5 = -15.9393$$

$$a_3 = -11.7811, \quad a_6 = +1.77516$$

$$p_{c,\text{H}_2\text{O}} = \text{critical pressure of H}_2\text{O} (22.064 \times 10^6 \text{ Pa})$$

$$T_{c,\text{H}_2\text{O}} = \text{critical temperature of H}_2\text{O} (647.14 \text{ K})$$

and where

$$\tau = 1 - \frac{T}{T_{c,\text{H}_2\text{O}}} \quad (\text{D3.4})$$

Calculation of Molar Volume of H₂O at Temperature T

The [Rowe and Chou \(1970\)](#) equation is used to calculate the specific volume of H₂O:

$$\hat{v}_{\text{H}_2\text{O}} = A(T) - \hat{p} B(T) - \hat{p}^2 C(T) \quad (\text{D3.5})$$

where

$$\hat{v}_{\text{H}_2\text{O}} = \text{specific volume of H}_2\text{O (cm}^3\text{/g)}$$

$$\hat{p} = \text{absolute pressure (kg/cm}^2\text{)}$$

The other parameters in Equation (D3.5) are given by:

$$\mathbf{A}(T) = 5.916365 - 0.01035794T + 0.9270048 \times 10^{-5}T^2 - 1127.522/T + 100674.1/T^2 \quad (\text{D3.6})$$

$$\begin{aligned} \mathbf{B}(T) = & 0.5204914 \times 10^{-2} - 0.1048210 \times 10^{-4}T + 0.8328532 \times 10^{-8}T^2 \\ & - 1.1702939/T + 1022783/T^2 \end{aligned} \quad (\text{D3.7})$$

$$\mathbf{C}(T) = 0.118547 \times 10^{-7} - 0.6599143 \times 10^{-10}T \quad (\text{D3.8})$$

The molar volume of H₂O (m³/mol) is then given by

$$v_{\text{H}_2\text{O}} = 10^{-6} \hat{v}_{\text{H}_2\text{O}} \times M_{\text{H}_2\text{O}} \quad (\text{D3.9})$$

where $M_{\text{H}_2\text{O}} = 18.015$ is the molecular weight of H₂O.

Calculation of Fugacity of Pure H₂O at Pressure p and Temperature T

The fugacity of pure H₂O at p and T is calculated from:

$$f_{\text{H}_2\text{O}} = f_{\text{H}_2\text{O}}^s \exp \left(\int_{p_{\text{H}_2\text{O}}^s}^p \frac{v_{\text{H}_2\text{O}}}{RT} dp \right) \quad (\text{D3.10})$$

where $f_{\text{H}_2\text{O}}^s$ and $v_{\text{H}_2\text{O}}$ are estimated from Equations [\(D3.2\)](#) and [\(D3.9\)](#), respectively.

Calculation of Fugacity of H₂O at p and in T Aqueous Solution

Finally the fugacity of H₂O in the aqueous solution is calculated from:

$$f_{\text{H}_2\text{O},w} = y_{\text{H}_2\text{O},w} f_{\text{H}_2\text{O}} \quad (\text{D3.11})$$

where $y_{\text{H}_2\text{O},w}$ is the mole fraction of H₂O in the aqueous phase.